



InPlay NanoBeacon™ IN100 Command Tool Application Notes



About Documentation

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This document applies to the following products:

Document number	Applicable Products	Document Status
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	IN100-Q1-R-RC1I	Mass production
	IN100-D1-R-YC1 I	Mass production
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	IN100-Q1-R-YC1F	Mass production

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1 Introduction

The NanoBeacon IN100 MP (Mass Production) command tool is used to program and test the IN100 device using the command line mode. The IN100 device supports a host controller interface, as shown in Figure 1, and this document will describe how to program and test the device with a host controller through this interface.

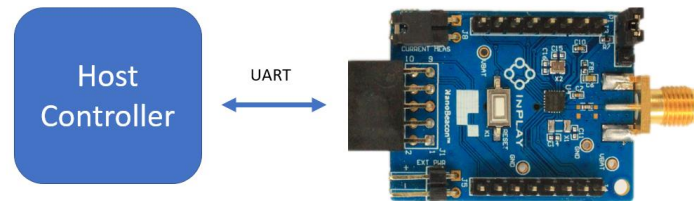


Figure 1: NanoBeacon device in host controller mode

The MP Command Tool is used with NanoBeacon ConfigTool to facilitate the user to perform eFuse programming and test during mass production.

The process is as follows.

1. First get the configuration with NanoBeacon ConfigTool according to the application requirements. Then save the configuration to a file (the config file).
2. Use the config file saved in step 1 as input and run MP command line tool(beacon_mp_cmd.exe)to burn the configuration into the eFuse memory of the IN100 device.
3. The operation flow of the MP command tool is shown in Figure 2.

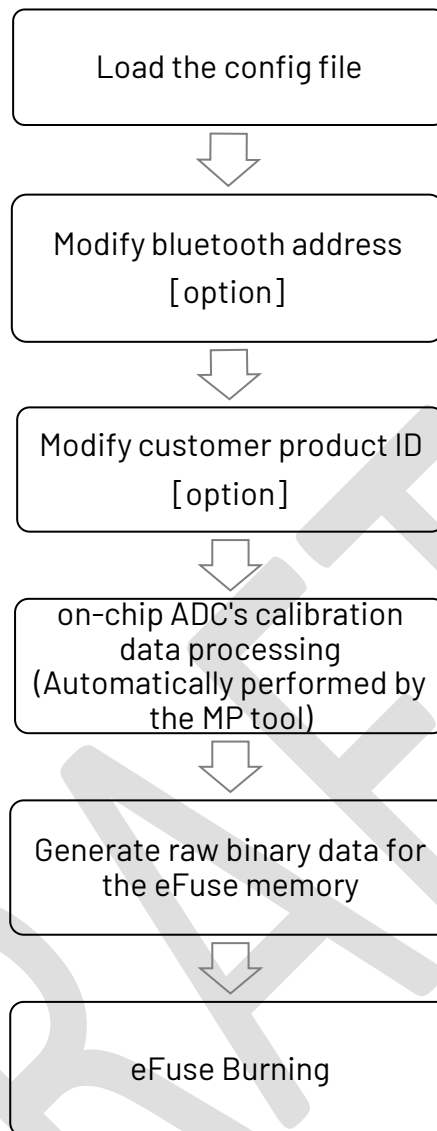


Figure 2: Operation flow of MP command tool

2 Commands

Commands are entered into the command line by typing the command arguments, followed by the value for the command. Some arguments do not require values after the argument. For example, the help command which uses the argument ‘-h’ or ‘--help’ need no further entry. Other arguments, like the set UART port, need a value after the argument.

The command line tool path and configuration file path should not be too deep!

2.1 eFuse burning

2.1.1 eFuse burning command

eFuse Burning refers to the programming of the IN100 OTP (One-Time Programming) eFuse memory. There is a 4Kbit eFuse memory which stores the configuration and advertising data packet information. The eFuse burning is permanent: Once the eFuse memory is burned, it is irreversible. This step (or this command) should be the last step performed after thoroughly testing or checking the configuration file.

The eFuse burning command can be used to burn eFuse for one or multiple IN100 chips. Table 1 lists the arguments for that command. The command arguments of type “Mandatory” are required arguments for the burning command to execute. For example, the UART port and the config file must be entered using the ‘-p’ and ‘-i’ arguments.

Table 1: The eFuse burning command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c burn	Command to burn eFuse memory	Mandatory
-p, --uart-port	beacon_mp_cmd.exe -c burn -p COM4	Set the UART port to be used/	Mandatory
-i, --src-file	beacon_mp_cmd.exe -c burn -p COM4 -i x.cfg	Set the config file to be used as input file (The config file comes from NanoBeacon config tool).	Mandatory
-a, --address	beacon_mp_cmd.exe -c burn -p COM4 -i x.cfg -a 0605AE030201	Using the argument to update Bluetooth device address. Six bytes of hexadecimal data, MSB first. Addresses of multiple advertising sets are separated by semicolons (;).	Optional
-d, --product-id	beacon_mp_cmd.exe -c burn -p COM4 -i x.cfg -d 0000000000F1	Using the argument to update customer product ID. Six bytes of hexadecimal data, MSB first.	Optional
-r, --reset-chip	beacon_mp_cmd.exe -c burn -p COM4 -r 1 -i x.cfg	1: Reset chip after finishing eFuse burning; 0: Not reset chip after burning	Optional

2.1.2 eFuse burning examples

2.1.2.1 Example 1

Burn the device using a config file named "nano_beacon_config.cfg". The UART port is "COM4".

```
beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg
```

2.1.2.2 Example 2

Burn the device using a config file named "nano_beacon_config.cfg" with BLE device address 0x0605AE030201. The UART port is "COM4" (Be noted if there are multiple advertising sets, only the device address of the first set will be modified with the provided address in the command line. The other sets will still use the addresses inside the configuration file).

```
beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg -a 0605AE030201
```

If there are two advertising sets, and we want to modify the device addresses for both of the sets, we can use the following command line (Be noted that if there are three advertising sets, only the device addresses of the first two sets will be modified):

```
beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg -a 0605AE030201; 0605AE030202
```

If there are three advertising sets, and we want to modify the device addresses for all of the three sets, we can use the following command line:

```
beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg -a 0605AE030201; 0605AE030202;  
0605AE030203
```

2.1.2.3 Example 3

Modify the Customer Product ID. The customer product ID is a user defined value, typically 4 – 6 bytes, that can be used like a serial number and added to a payload. All advertising sets share the same Customer Product ID:

```
beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg -d 0000000000F1
```

2.1.3 eFuse burning log

2.1.3.1 Successful burning

After the MP tool finishes the eFuse programming, it will read out the eFuse content from the IN100 device and compare with the content from the config file and/or the device addresses/product ID provided in the command line. If the readouts are the expected, the command line process returns 0, and the last line of log is "ok burn", as shown in Figure 3.

```
D:\NanoBeaconConfigToolMpTool_V3.2.25\NanoBeaconTools>beacon_mp_cmd.exe -c burn -p COM42 -i logs/beacon_cfg.cfg
temp cal eFuse = 65469, unitLsb = 0.01
temp cal result: d040442,10934,1397350e
high_temp_flag
Num fields: 2
Num fields: 3
Num fields: 4
Num fields: 5
Warning: bigger size than command line value (82): 132
cs header last byte idx: 15 last value: 65_53_07
Set pkt offset to: 1b with CS offset 6
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
Set pkt offset to: 1b with CS offset 6
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
[74, 206]
[206, 206]
132
Region 1 address range: [16, 73]
Region 2 address range: [74, 206]
Region 3 address range: [207, 232]
ok burn
```

Figure 3: Successful eFuse burning

2.1.3.2 Failed burning

If the eFuse burning (the readouts are not expected) does not go through, the command line process returns -1, and the last line of log will be "error xxx", as shown in Figure 4.

```
D:\NanoBeaconConfigToolMpTool_V3.2.25\NanoBeaconTools>beacon_mp_cmd.exe -c burn -p COM42 -i logs/beacon_cfg.cfg
temp cal eFuse = 65469, unitLsb = 0.01
temp cal result: d040442,10934,1397350e
high_temp_flag
Num fields: 2
Num fields: 3
Num fields: 4
Num fields: 5
Warning: bigger size than command line value (82): 132
cs header last byte idx: 15 last value: 65_53_07
Set pkt offset to: 1b with CS offset 6
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
Set pkt offset to: 1b with CS offset 6
cs header last byte idx: 15 last value: 65_53_07
cs header last byte idx: 15 last value: 65_53_07
[74, 206]
[206, 206]
132
Region 1 address range: [16, 73]
Region 2 address range: [74, 206]
Region 3 address range: [207, 232]
error burn failed eFuse 10 write 6300, read back 0000
```

Figure 4: Failed eFuse burning

2.2 RAM test

2.2.1 RAM test command (Running-in-RAM test command)

When 'Running-in-RAM,' the MP Tool issues UART commands to the IN100 device to make it run the config file without burning the eFuse memory. This is not permanent, so power cycling or resetting the IN100 will end the RAM test. The purpose of the RAM test is to test if the settings in the config file will work as needed for an application without wasting a chip by burning eFuse. Please be noted, I2C is not supported for the "Running-in-RAM" mode. The details of the RAM test command is shown in Table 2.

Table 2: The RAM test command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c ram	Command to start RAM test	Mandatory
-p, --uart-port	beacon_mp_cmd.exe -c ram -p COM4	Which UART port to use	Mandatory
-i, --src-file	beacon_mp_cmd.exe -c ram -p COM4 -i x.cfg	Which config file to be used as the input file. The config file comes from NanoBeacon config tool.	Mandatory
-a, --address	beacon_mp_cmd.exe -c ram -p COM4 -i x.cfg -a 0605AE030201	Using the argument to update Bluetooth device address. Six bytes of hexadecimal data, MSB first. Addresses of multiple advertising sets are separated by semicolons (;).	Optional
-d, --product-id	beacon_mp_cmd.exe -c ram -p COM4 -i x.cfg -d 0000000000F1	Using the argument to update customer product ID. Six bytes of hexadecimal data, MSB first.	Optional

2.2.2 RAM test examples

2.2.2.1 Example 1

Run the RAM test for the device using a config file called "nano_beacon_config.cfg"

beacon_mp_cmd.exe -c burn -p COM4 -i nano_beacon_config.cfg

2.2.2.2 Example 2

Run the RAM test for the device using a config file named "nano_beacon_config.cfg" with BLE device address 0x0605AE030201. The UART port is "COM4" (Be noted if there are multiple advertising sets, only the device address of the first set will be modified with the provided address in the command line. The other sets will still use the addresses inside the configuration file).

beacon_mp_cmd.exe -c ram -p COM4 -i nano_beacon_config.cfg -a 0605AE030201

If there are two advertising sets, and we want to modify the device addresses for both of the sets, we can use the following command line (Be noted that if there are three advertising sets, only the device addresses of the first two sets will be modified):

```
beacon_mp_cmd.exe -c ram -p COM4 -i nano_beacon_config.cfg -a 0605AE030201; 0605AE030202
```

If there are three advertising sets, and we want to modify the device addresses for all of the three sets, we can use the following command line:

```
beacon_mp_cmd.exe -c ram -p COM4 -i nano_beacon_config.cfg -a 0605AE030201; 0605AE030202;  
0605AE030203
```

2.2.2.3 Example 3

Modify the Customer Product ID. The customer product ID is a user defined value, typically 4 – 6 bytes, that can be used like a serial number and added to a payload. All advertising sets share the same Customer Product ID:

```
beacon_mp_cmd.exe -c ram -p COM4 -i nano_beacon_config.cfg -d 0000000000F1
```

2.3 Bluetooth DTM test

Direct Test Mode (DTM) is used to let the device send out DTM test packets, specified by the Bluetooth core specification. For more information on the technical details and purpose of DTM packets, please check out the Bluetooth Core Specification 5.3.

DTM can be performed over HCI or 2-wire UART interface as defined in the Bluetooth 5.3 Core Specification. The IN100 supports DTM through the 2-wire UART method which can be used either with the command line tool or the NanoBeacon ConfigTool. DTM settings include the configurations of the RF channel index, payload length, payload pattern, and the PHY rate. Additionally, the user can also configure the XO crystal cap setting and Tx power level.

The IN100 device provides programmable load capacitance for the external 26MHz crystal. The cap setting, also called the 'Cap Code', configures the value of the on-chip capacitance for the crystal's load capacitance. The formula of cap code to CL (Load Capacitance) is as follows:

$$CL = (\text{capcode} + 1) / 2 \text{ (pF)}$$

The capcode should be configured according to the crystal specification and the user's PCB.

Tx power level refers to the power in dBm at which the RF signal can be transmitted. The maximum value is +4 and the minimum value is -54 dBm.

2.3.1 DTM test start command

To let the IN100 device send out DTM test packets, the user can issue DTM start command (details shown in Table 3) using the MP command line tool.

Table 3: The DTM start command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c dtm_start	Bluetooth DTM test start command	Mandatory
-p, --uart-port	beacon_mp_cmd.exe -c dtm_start -p COM4	Which UART port to use	Mandatory
-x, --parameter	beacon_mp_cmd.exe -c dtm_start -p COM4 -x 37,25,0,1M,11,3	<ch,tx_len,tx_payload,phy,cap,tx_power> Parameters to be set for DTM test. There are six parameters in total, separated by commas (.). ch: physical channel 0~39; tx_len: length for transmitting 0 ~ 255; tx_payload: index of payload pattern 0~7 0 - PRBS9 1 - 0x0F	Mandatory

Arguments	Example	Description	Type
		2 - 0x55 3 - PRBS15 4 - 0xFF 5 - 0x0 6 - 0xF0 7 - 0xAA phy: '1M' for 1M PHY, '125K' for 125K coded PHY cap: on-chip capacitor code 0~15 tx_power: tx power, dBm	

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2.3.2 DTM test stop command

DTM stop command is used to stop the DTM test.

Table 4: The DTM stop command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c dtm_stop	DTM test stop command	Mandatory
-p, --uart-port	beacon_mp_cmd.exe -c dtm_stop -p COM4	Which UART port to use	Mandatory

2.3.3 DTM Example

2.3.3.1 DTM start test command

The following command will start the DTM test using UART port 'COM4.' It will transmit on RF channel 1 with a payload length of 37. It will use payload patterns PBRS9 at a 1 M PHY rate and cap code of 11 and at 3 dBm tx power:

beacon_mp_cmd.exe -c dtm_start -p COM4 -x 1,37,0,1M,11,3

2.3.3.2 DTM stop test command

This command will stop the DTM test.

beacon_mp_cmd.exe -c dtm_stop

2.4 Carrier test

The carrier test let the IN100 device transmit an unmodulated continuous waveform (CW) on a selected channel. Like the DTM test, the user needs to configure the crystal's load capacitance and TX power settings.

2.4.1 Carrier test start command

To let the IN100 send out a carrier tone, the user needs to issue a carrier start command specified in Table 5.

Table 5: The carrier test start command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c carrier_start	Start carrier test command	Mandatory
-p, --uart-port	beacon_mp_cmd.exe -c carrier_start -p COM4	Which UART port to use	Mandatory
-x, --parameter	beacon_mp_cmd.exe -c carrier_start -p COM4 -x 37,11,3	<ch,cap,tx_power> Parameters to be set for carrier test. There are three parameters in total, separated by commas (,). ch: physical channel 0~39; cap: on-chip capacitor code 0~15 tx_power: tx power, dBm	Mandatory

2.4.2 Carrier test stop command

To stop the carrier test, the user needs to issue a stop command specified in Table 6.

Table 6: The carrier test stop command and its arguments

Arguments	Example	Description	Type
-c, --command-name	beacon_mp_cmd.exe -c carrier_stop	Stop carrier test command	Mandatory
-x, --parameter	beacon_mp_cmd.exe -c carrier_stop -p COM4	Which UART port to use	Mandatory

2.4.3 Carrier test examples

2.4.3.1 Carrier test start

This command will start a carrier test on UART port 'COM4' on channel 37 with a cap code of 11 and a Tx power level of +3 dBm:

beacon_mp_cmd.exe -c carrier_start -p COM4 -x 37,11,3

2.4.3.2 Carrier test stop

This command will stop the carrier test on UART port 'COM4':

beacon_mp_cmd.exe -c carrier_stop -p COM4

3 Revision History

Revision	Description	Prepared By	Date
V0.9	Preliminary version	Aidan Martens	12/08/2023

4 Disclaimer

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